**QPCode** **Reg.No : .....**

**Name : .....**

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) Exam Type EXAMINATION Month YYYY**

**Semester 4**

**VAC -** MG4VACPHYA00**- MEASUREMENT AND PROCESS CONTROL**

(2025 ADMISSION ONWARDS)

**Duration: 2 hour** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

**Part A**

Short Answer

Answer any **7** questions

Each question carries **2** marks

| | | |
|---|---|---|
| 1. | Define instrumentation. | [K] / [1] |
| 2. | Distinguish between absolute instruments and secondary instruments. | [U] / [1] |
| 3. | Explain direct method of measurement. Give an example. | [U] / [2] |
| 4. | What is the function of a primary sensing element? | [U] / [2] |
| 5. | A laboratory experiment involves measuring high-pressure gas flow. Name a suitable instrument for this purpose and classify it based on its mode of operation. | [A] / [3] |
| 6. | A digital thermometer is used in a laboratory experiment. Classify the instrument based on its mode of operation and justify your classification. | [A] / [3] |
| 7. | Define a transducer. | [K] / [4] |
| 8. | List any two static characteristics of transducers. | [U] / [4] |
| 9. | What is an RTD? | [K] / [5] |
| 10. | State the principle of a strain gauge. | [U] / [5] |

| | | |
|---|---|---|
| 11. | Define process control system. | [K] / [6] |
| 12. | What is a Programmable Logic Controller (PLC)? | [K] / [6] |

**[2x7 = 14]**

**Part B**

Short Essay

Answer any **6** questions

Each question carries **5** marks

| | | |
|---|---|---|
| 13. | Define Measurement. Explain conditions for obtaining a meaningful measurement. | [U] / [1] |
| 14. | A physics laboratory uses different measurement systems for electrical and mechanical experiments. Identify suitable measurement systems for each case and explain how the choice of measurement system affects the accuracy and reliability of measurements. | [A] / [1] |
| 15. | Describe the elements of a measurement system with a neat block diagram. | [U] / [2] |
| 16. | Explain the classification of instruments based on mode of operation. | [U] / [3] |
| 17. | Explain various classifications of instruments. | [U] / [3] |
| 18. | Classify transducers based on principle of transduction and explain any two. | [E] / [4] |
| 19. | Explain the working principle and applications of strain gauges. | [A] / [5] |
| 20. | Describe RTD and thermistors, analyse their advantages and applications. | [An] / [5] |
| 21. | Explain control systems and controllers used in process control. | [An] / [6] |
| 22. | Explain the construction and working of LVDT with applications. | [A] / [5] |
| 23. | How do modern digital sensors and transducers make measurement systems more attractive and user-friendly compared to traditional instruments? | [I] / [5] |
| 24. | Explain control systems and controllers used in process control. | [An] / [6] |

**[6x6 = 36]**